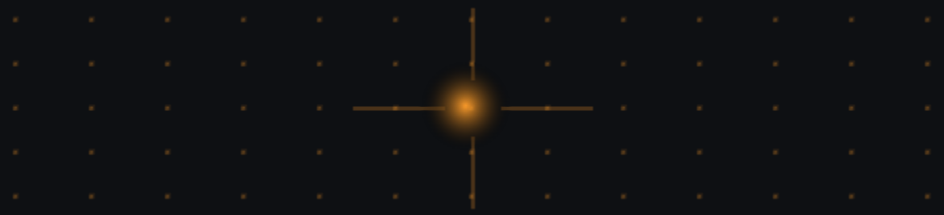


AaltoCamera



Laser-spot tracking, sample stabilisation and autofocus.

In every frame the module finds the fixed laser spot and a pattern on the sample, and places a grid of measurement points on that pattern. The stabiliser moves the sample, through the stage module, so that the selected point stays under the laser spot. Imaging and spot detection run on the lab's camera; the stabiliser and autofocus have so far been tested in simulation.

IDS uEye+ camera (IDS peak) or a GenICam camera · XY and Z through the **KIM101** stage, or a **piezo** scanner with a **KCube** Z piezo

Module source on GitHub

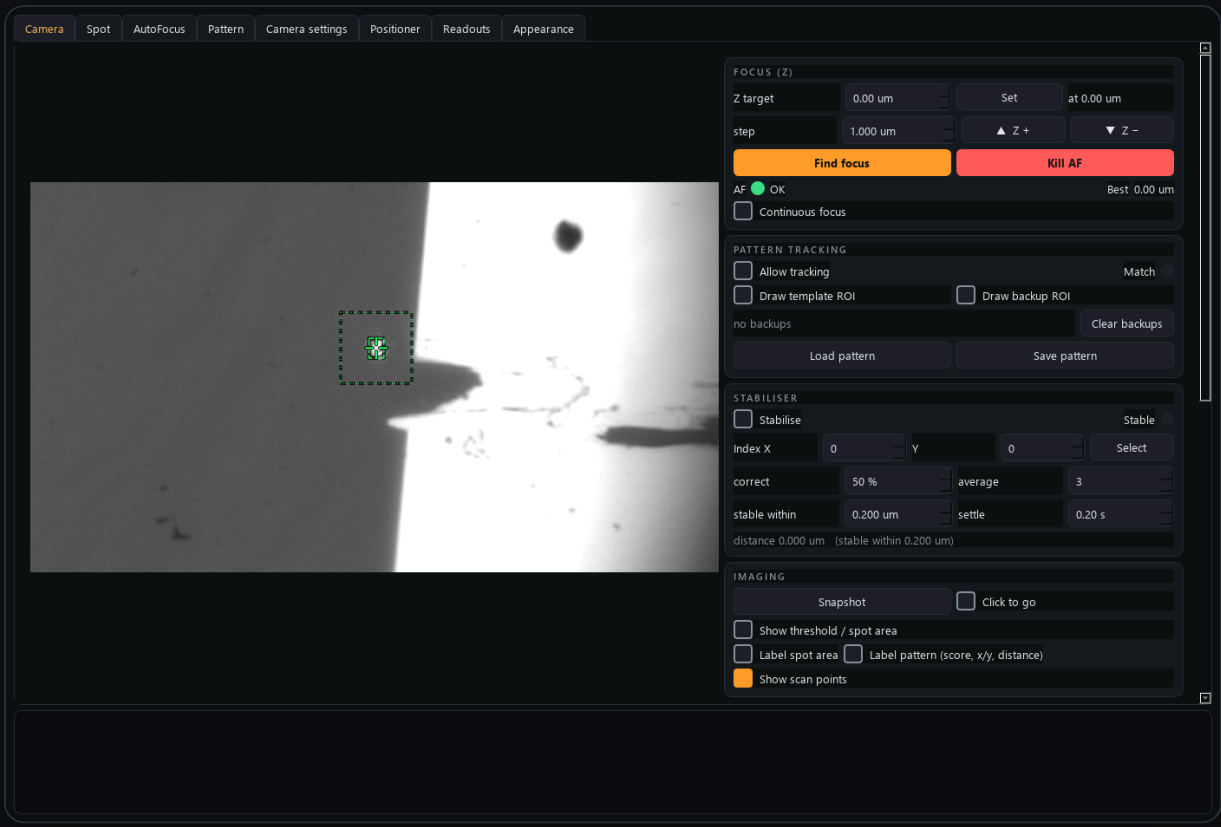
AaltoFlow on GitHub

pattern tracking stabiliser autofocus click to go simulator included

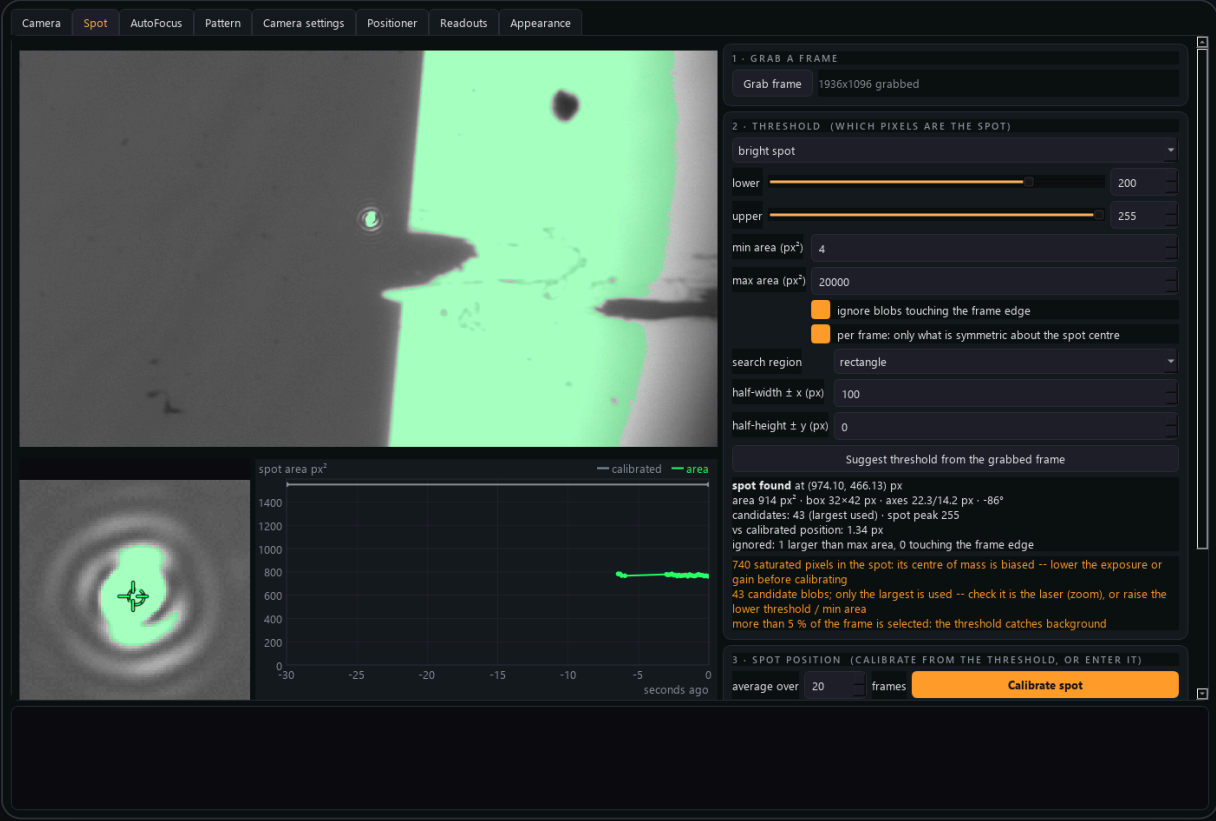
IMAGE

Camera view

The live image with the calibrated laser spot marked. Clicking in the image moves the sample so that the clicked point goes under the spot.



On the lab setup, 63x objective: the laser spot (marked) at the edge of a structure



Spot tab on the lab setup: threshold, zoomed spot, spot size over time

CALIBRATE

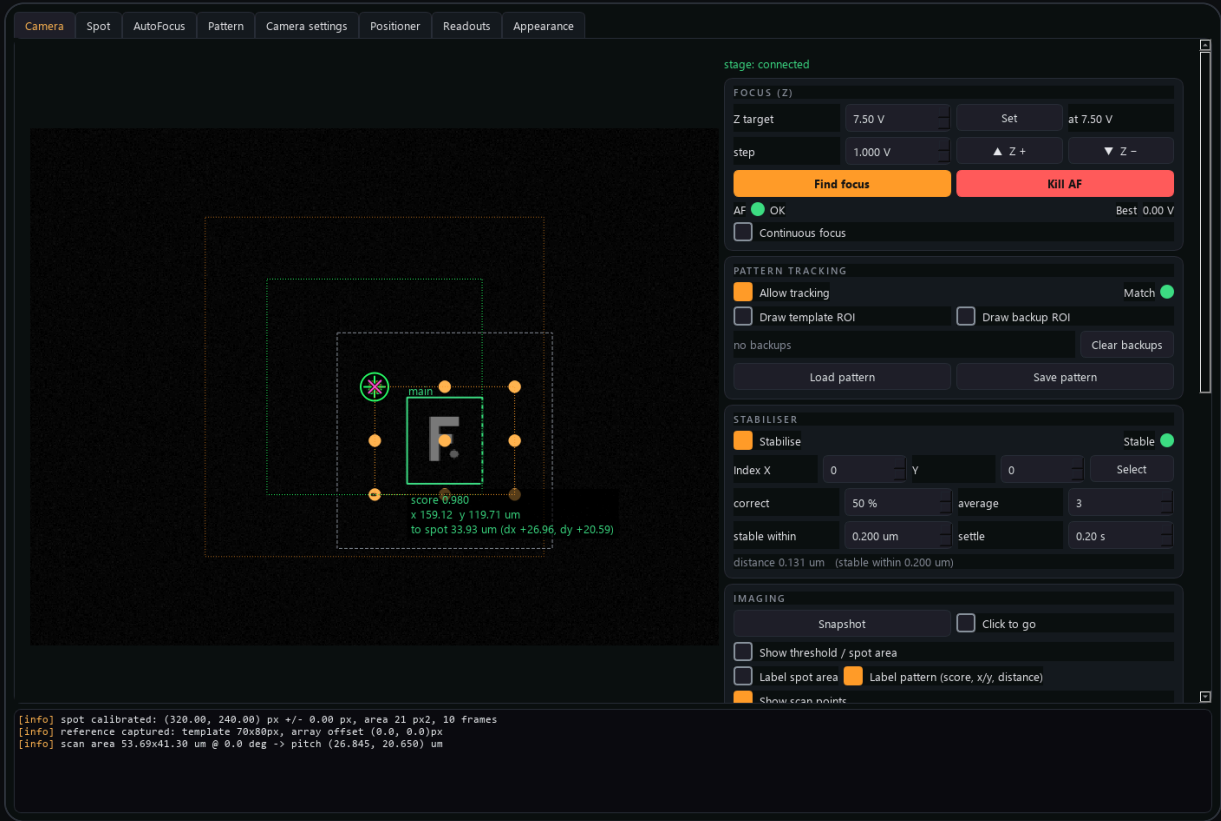
Spot calibration

The spot detection is set on a grabbed frame: threshold, search region, a zoomed view and the spot size over time. The calibrated spot position is used by click-to-go and by the stabiliser.

STABILISE

Pattern, scan array and stabiliser

A template drawn on the sample is found in every frame (OpenCV, sub-pixel). The scan array is placed relative to it, so it moves with the sample. The stabiliser averages a set number of frames and then moves the sample by a set fraction of the error, until the selected array point is within a set distance of the spot.



Simulation: tracked pattern (match score 0.98), 3 × 3 scan array, point (0, 0) held under the spot

Autofocus

A Z sweep with a fit, or a one-direction search for hysteretic slip-stick Z. It can run as a scan routine; the scan waits for it.

Scan array as scan axes

In AaltoFlow the array index is a scan axis: the stabiliser brings each point under the laser before it is measured.

Backup patterns

When the main pattern leaves the field of view, a backup pattern keeps the array in place.

Camera settings

Exposure, gain, ROI and the other parameters are read from the camera's own feature list.

Saving

The pattern file contains the scan-array settings. As scan routines, the pattern and a camera picture are saved into the measurement's folder before and after a scan.

Objectives

Choosing the objective sets the pixel size, from an editable table.

PART OF AALTOFLOW

AaltoFlow

In AaltoFlow each instrument runs as its own service with its own window. A launcher starts them, and a scan engine drives any combination of them through multi-dimensional measurements.

github.com/FlashLukas/AaltoFlow

[camera-control](#) on GitHub

LICENCE AND CITATION

Citation

Released under the MIT licence. If you publish data measured with it, please cite:

L. Flajšman, *AaltoFlow: lab automation built from independent instrument modules*, NanoSpin group, Aalto University, doi:10.5281/zenodo.22959230

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